Concentrated atom-adic eigenvalue valuations for the staircase operator on $V_{(4,3)}$ and $V_{(5,2)}$ at $n = 7$

Clio

28 April 2026

Abstract

Let $\Pi_q \in H_q(\mathfrak{S}_7)$ be the staircase Bott–Samelson product $\prod_{j=1}^{21}(1+T_{i_j})$ along $\underline{w}_0 = 1\cdot 21\cdot 321\cdot 4321\cdot 54321\cdot 654321$, and let $\mathfrak{a}(q) = q^2+4q+1 \in \mathbb{Z}[q]$. Acting on the Specht modules $V_{(4,3)}$ and $V_{(5,2)}$, Π_q has rank 4 and 5 respectively. We prove that, in any extension of $\mathbb{Q}(q)$ where the characteristic polynomial of $\Pi_q|_{V_\lambda}$ splits, the non-zero eigenvalues satisfy

$$V_{(4,3)/n=7}: \quad \nu_{\text{atom}}(e_1) = 3, \quad \nu_{\text{atom}}(e_2) = \nu_{\text{atom}}(e_3) = \nu_{\text{atom}}(e_4) = 0,$$
$$V_{(5,2)/n=7}: \quad \nu_{\text{atom}}(e_1) = 3, \quad \nu_{\text{atom}}(e_2) = \nu_{\text{atom}}(e_3) = \nu_{\text{atom}}(e_4) = \nu_{\text{atom}}(e_5) = 0,$$

where ν_{atom} is the $\mathfrak{a}$-adic valuation extended to the algebraic closure. We call this the *concentrated* distribution, in contrast to the *graded* distribution $(2,1,0)$ recently proved at $V_{(4,2)/n=6}$. The proof has two ingredients: (i) a symbolic computation, via Lagrange interpolation over $\mathbb{Q}$, of the elementary symmetric functions $\sigma_1,\dots,\sigma_r$ of the eigenvalues (with their explicit $\mathfrak{a}$-adic factorisations); and (ii) the Newton polygon theorem applied to the rank-r characteristic polynomial.

1 Setup

Let $H_q = H_q(\mathfrak{S}_n)$ be the Hecke algebra of the symmetric group with parameter q, with generators $T_1,\dots,T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid and commutation relations. For $n = 7$ fix the staircase reduced word

$$\underline{w}_0 = (i_1,\dots,i_{21}) = (1,\ \ 2,1,\ \ 3,2,1,\ \ 4,3,2,1,\ \ 5,4,3,2,1,\ \ 6,5,4,3,2,1)$$

for the longest element $w_0 \in \mathfrak{S}_7$, and let

$$\Pi_q \ = \ (1+T_{i_1})(1+T_{i_2})\cdots(1+T_{i_{21}}) \ \in \ H_q.$$

For each partition $\lambda \vdash n$, write V_λ for the cell (Specht) module of dimension f^λ. We work throughout in Hoefsmit's seminormal form, where T_i acts by an explicit matrix over $\mathbb{Q}(q)$ in the basis indexed by standard Young tableaux [1].

We focus on $\lambda \in \{(4,3),(5,2)\}$ and $n = 7$. Then $f^{(4,3)} = f^{(5,2)} = 14$, so $\Pi_q|_{V_\lambda}$ is in each case a 14×14 matrix with entries in $\mathbb{Q}(q)$.

Let $\mathfrak{a} = \mathfrak{a}(q) := q^2+4q+1 \in \mathbb{Z}[q]$. This polynomial is irreducible over $\mathbb{Q}$ (its roots are $-2\pm\sqrt{3}$, irrational) and primitive. The ideal $(\mathfrak{a})$ is therefore a height-1 prime in the principal ideal domain $\mathbb{Q}[q]$, and the localisation $\mathbb{Q}[q]_{(\mathfrak{a})}$ is a discrete valuation ring with uniformiser $\mathfrak{a}$. Write

$$\nu_{\text{atom}}\colon \mathbb{Q}(q)^\times \longrightarrow \mathbb{Z}$$

for the resulting $\mathfrak{a}$-adic valuation: for a non-zero polynomial $f \in \mathbb{Q}[q]$, $\nu_{\text{atom}}(f)$ is the largest integer m with $\mathfrak{a}^m \mid f$, and ν_{atom} extends uniquely to $\mathbb{Q}(q)$ by $\nu_{\text{atom}}(f/g) = \nu_{\text{atom}}(f) - \nu_{\text{atom}}(g)$.

Definition 1. For $k \geq 1$ define $\sigma_k(\Pi_q|_{V_\lambda}) \in \mathbb{Q}(q)$ as the k-th elementary symmetric polynomial of the eigenvalues of $\Pi_q|_{V_\lambda}$ (counted with multiplicity in any algebraic-closure extension), equivalently the trace of the k-th exterior power $\Lambda^k(\Pi_q|_{V_\lambda})$. These are computed concretely by Newton's identities from $p_i := \mathrm{tr}((\Pi_q|_{V_\lambda})^i)$:

$$\sigma_k = \frac{1}{k}\sum_{i=1}^{k}(-1)^{i-1}\sigma_{k-i}\, p_i, \qquad \sigma_0 = 1.$$

2 Main theorems

Theorem 2 (T2)**.** *Let $g(t) \in \mathbb{Q}(q)[t]$ be the monic minimal polynomial vanishing on the non-zero eigenvalues of $\Pi_q|_{V_{(4,3)/n=7}}$, equivalently the degree-4 factor of the 14×14 characteristic polynomial after stripping the kernel factor t^{10}. Extend the $\mathfrak{a}$-adic valuation ν_{atom} to the algebraic closure of $\mathbb{Q}(q)$. Then $g(t)$ is monic of degree 4, and its four roots e_1, e_2, e_3, e_4 satisfy*

$$\nu_{\mathrm{atom}}(e_1) = 3, \qquad \nu_{\mathrm{atom}}(e_2) = \nu_{\mathrm{atom}}(e_3) = \nu_{\mathrm{atom}}(e_4) = 0.$$

Theorem 3 (T3)**.** *Let $g(t) \in \mathbb{Q}(q)[t]$ be the monic minimal polynomial vanishing on the non-zero eigenvalues of $\Pi_q|_{V_{(5,2)/n=7}}$, equivalently the degree-5 factor of the 14×14 characteristic polynomial after stripping the kernel factor t^9. Extend the $\mathfrak{a}$-adic valuation ν_{atom} to the algebraic closure of $\mathbb{Q}(q)$. Then $g(t)$ is monic of degree 5, and its five roots $e_1, \ldots, e_5$ satisfy*

$$\nu_{\mathrm{atom}}(e_1) = 3, \qquad \nu_{\mathrm{atom}}(e_2) = \nu_{\mathrm{atom}}(e_3) = \nu_{\mathrm{atom}}(e_4) = \nu_{\mathrm{atom}}(e_5) = 0.$$

The proofs of T2 and T3 follow the same template: symbolically pin down $\sigma_1, \ldots, \sigma_r$, verify $\sigma_{r+1} \equiv 0$ to confirm the rank, and apply the Newton polygon theorem.

3 Symbolic computation: σ_k for $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$

Lemma 4. *For $\lambda = (4,3)$, $n = 7$, the elementary symmetric polynomials $\sigma_k = \sigma_k(\Pi_q|_{V_{(4,3)/n=7}}) \in \mathbb{Z}[q]$ satisfy*

$$\begin{aligned}
\sigma_1 &= q^3(1+q)^3 \cdot P_1(q), && (1)\\
\sigma_2 &= q^7(1+q)^{10} \cdot P_2(q), && (2)\\
\sigma_3 &= q^{12}(1+q)^{21} \cdot P_3(q), && (3)\\
\sigma_4 &= q^{18}(1+q)^{36} \cdot \mathfrak{a}(q)^3 \cdot A_9(q), && (4)\\
\sigma_5 &= 0, && (5)
\end{aligned}$$

where $A_9(q) = q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$ and P_1, P_2, P_3 are explicit palindromic polynomials of degrees $12, 18, 18$ respectively, all coprime to $\mathfrak{a}(q)$ (see appendix for explicit coefficient lists). Consequently

$$\nu_{\mathrm{atom}}(\sigma_1) = 0, \quad \nu_{\mathrm{atom}}(\sigma_2) = 0, \quad \nu_{\mathrm{atom}}(\sigma_3) = 0, \quad \nu_{\mathrm{atom}}(\sigma_4) = 3.$$

Lemma 5. *For $\lambda = (5,2)$, $n = 7$, the elementary symmetric polynomials $\sigma_k = \sigma_k(\Pi_q|_{V_{(5,2)/n=7}}) \in \mathbb{Z}[q]$ satisfy*

$$\sigma_1 = q^2(1+q)^5 \cdot Q_1(q), \tag{6}$$
$$\sigma_2 = q^5(1+q)^{14} \cdot Q_2(q), \tag{7}$$
$$\sigma_3 = q^9(1+q)^{27} \cdot Q_3(q), \tag{8}$$
$$\sigma_4 = q^{13}(1+q)^{40} \cdot Q_4(q), \tag{9}$$
$$\sigma_5 = q^{18}(1+q)^{57} \cdot \mathfrak{a}(q)^3 \cdot A_9(q), \tag{10}$$
$$\sigma_6 = 0, \tag{11}$$

where $A_9(q) = q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$ and Q_1, Q_2, Q_3, Q_4 are explicit palindromic polynomials of degrees $12, 18, 18, 18$ respectively, all coprime to $\mathfrak{a}(q)$. Consequently

$$\nu_{\text{atom}}(\sigma_1) = \nu_{\text{atom}}(\sigma_2) = \nu_{\text{atom}}(\sigma_3) = \nu_{\text{atom}}(\sigma_4) = 0, \quad \nu_{\text{atom}}(\sigma_5) = 3.$$

Proof of Lemmas 4 and 5. The proof is computational, but rigorous: it is exact rational arithmetic, not numerics.

Let $P^{(\nu)} \in \mathrm{Mat}_{14}(\mathbb{Q})$ denote the matrix obtained from $\Pi_q|_{V_\lambda}$ by specialising q to a positive rational integer $\nu \geq 2$. The Hoefsmit denominators $1 - q^{-d}$ for axial-distance integers d that occur in the shape do not vanish at any integer $\nu \geq 2$, so $P^{(\nu)}$ is well-defined as a matrix in $\mathbb{Q}$.

The map $\nu \mapsto \sigma_k(P^{(\nu)})$ takes integer values that agree with the polynomial $\sigma_k(q) \in \mathbb{Z}[q]$ evaluated at $q = \nu$. The polynomial $\sigma_k(q)$ has q-degree bounded by $k \cdot 21$ (each of the 21 factors $1 + T_{i_j}$ has matrix entries of q-degree at most 1, so entries of Π_q have q-degree at most 21, and σ_k as a sum of principal $k \times k$ minors has q-degree at most $21k$).

For $\lambda = (4,3)$: we evaluate $\sigma_k(P^{(\nu)})$ at $\nu = 2, 3, \ldots, 90$ (89 points), exceeding the degree bound $4 \cdot 21 = 84$. Lagrange interpolation over $\mathbb{Q}$ then recovers the unique $\sigma_k(q)$ of degree ≤ 84 matching all 89 evaluations. Overdetermination by 5 points provides a built-in cross-check.

For $\lambda = (5,2)$: we evaluate at $\nu = 2, \ldots, 111$ (110 points), exceeding the degree bound $5 \cdot 21 = 105$.

The recovered factorisations are exactly as stated in the lemmas, and were verified by independent computation in two separate scripts (`compute_v3.py` and `verify.py`).

Coprimality of each P_k, Q_k to $\mathfrak{a}$ is verified symbolically: substituting $q = -2 + \sqrt{3}$ (a root of $\mathfrak{a}$ in $\mathbb{Q}(\sqrt{3})$) into each remainder polynomial gives a non-zero element of $\mathbb{Q}(\sqrt{3})$ in every case (see Appendix A).

Vanishing of σ_5 (resp. σ_6) is verified by direct evaluation: at all 89 (resp. 110) integer points the result is exactly $0 \in \mathbb{Q}$, which exceeds the degree bound, so the polynomial is identically zero. □

Corollary 6. $\mathrm{rk}(\Pi_q|_{V_{(4,3)/n=7}}) = 4$ *and* $\mathrm{rk}(\Pi_q|_{V_{(5,2)/n=7}}) = 5$.

Proof. For any matrix A over $\mathbb{Q}(q)$ of size d, $\sigma_k(A) = \mathrm{tr}(\Lambda^k A) = 0$ for all $k > \mathrm{rk}(A)$, while $\sigma_{\mathrm{rk}(A)}(A) \neq 0$. By Lemmas 4–5, $\sigma_4 \neq 0 = \sigma_5$ for $V_{(4,3)/n=7}$ and $\sigma_5 \neq 0 = \sigma_6$ for $V_{(5,2)/n=7}$. □

4 Newton polygon

Theorem 7 (Newton polygon, e.g. [2, Ch. II, Prop. 6.3])**.** *Let K be a field with a discrete valuation $v\colon K^\times \to \mathbb{Z}$, let $\bar{K}$ be an algebraic closure with v extended to $\bar{K}^\times \to \mathbb{Q}$. Let $f(t) = a_n t^n + a_{n-1} t^{n-1} + \cdots + a_0 \in K[t]$ with $a_n a_0 \neq 0$. The* Newton polygon *of f is the lower convex hull in $\mathbb{R}^2$ of the points $\{(i, v(a_i)) : 0 \leq i \leq n,\ a_i \neq 0\}$. If the polygon consists of edges of slopes $m_1 < m_2 < \cdots < m_r$ with*

respective horizontal lengths $\ell_1, \ldots, \ell_r$ ($\sum \ell_j = n$), then f has exactly ℓ_j roots in $\bar{K}$ (counted with multiplicity) whose valuation equals $-m_j$.

Proof of Theorem 2. By Corollary 6, the non-zero eigenvalues of $\Pi_q|_{V_{(4,3)/n=7}}$ are exactly the four roots in $\overline{\mathbb{Q}(q)}$ of

$$g(t) = t^4 - \sigma_1 t^3 + \sigma_2 t^2 - \sigma_3 t + \sigma_4.$$

We compute the $\mathfrak{a}$-adic valuations of the coefficients (indexed by power of t):

- $i = 4$ (leading): $v(1) = 0$.
- $i = 3$: $v(-\sigma_1) = v(\sigma_1) = 0$ by Lemma 4.
- $i = 2$: $v(\sigma_2) = 0$.
- $i = 1$: $v(-\sigma_3) = v(\sigma_3) = 0$.
- $i = 0$: $v(\sigma_4) = 3$.

The Newton polygon of g is the lower convex hull of the points

$$\big\{(0,3),\ (1,0),\ (2,0),\ (3,0),\ (4,0)\big\} \subset \mathbb{R}^2.$$

The hull consists of two segments:

$$\begin{array}{lll} (0,3) \to (1,0): & \text{slope } -3, & \text{horizontal length } 1. \\ (1,0) \to (4,0): & \text{slope } \ \ 0, & \text{horizontal length } 3. \end{array}$$

By Theorem 7, $g(t)$ has exactly one root of valuation $-(-3) = 3$ and exactly three roots of valuation $-0 = 0$. These four roots are the four non-zero eigenvalues e_1, e_2, e_3, e_4 of $\Pi_q|_{V_{(4,3)/n=7}}$. Relabelling so that $\nu_{\text{atom}}(e_1) \geq \nu_{\text{atom}}(e_2) \geq \nu_{\text{atom}}(e_3) \geq \nu_{\text{atom}}(e_4)$:

$$\nu_{\text{atom}}(e_1) = 3, \qquad \nu_{\text{atom}}(e_2) = \nu_{\text{atom}}(e_3) = \nu_{\text{atom}}(e_4) = 0. \qquad \square$$

Proof of Theorem 3. By Corollary 6, the non-zero eigenvalues of $\Pi_q|_{V_{(5,2)/n=7}}$ are exactly the five roots in $\overline{\mathbb{Q}(q)}$ of

$$g(t) = t^5 - \sigma_1 t^4 + \sigma_2 t^3 - \sigma_3 t^2 + \sigma_4 t - \sigma_5.$$

Coefficient atom-valuations: $0, 0, 0, 0, 0, 3$ at $i = 5, 4, 3, 2, 1, 0$ respectively, by Lemma 5. The Newton polygon is the lower convex hull of

$$\big\{(0,3),\ (1,0),\ (2,0),\ (3,0),\ (4,0),\ (5,0)\big\},$$

consisting of segments

$$\begin{array}{lll} (0,3) \to (1,0): & \text{slope } -3, & \text{horizontal length } 1. \\ (1,0) \to (5,0): & \text{slope } \ \ 0, & \text{horizontal length } 4. \end{array}$$

By Theorem 7, $g(t)$ has exactly one root of valuation 3 and four roots of valuation 0. Relabelling:

$$\nu_{\text{atom}}(e_1) = 3, \qquad \nu_{\text{atom}}(e_2) = \nu_{\text{atom}}(e_3) = \nu_{\text{atom}}(e_4) = \nu_{\text{atom}}(e_5) = 0. \qquad \square$$

5 Discussion

5.1 Concentrated vs. graded

The three known atom^3 shapes now have their atom-distribution decoded:

- $V_{(4,2)/n=6}$: **graded** $(2,1,0)$ (Theorem T1).
- $V_{(4,3)/n=7}$: **concentrated** $(3,0,0,0)$ (Theorem T2).
- $V_{(5,2)/n=7}$: **concentrated** $(3,0,0,0,0)$ (Theorem T3).

What selects between the two regimes is not yet understood. A natural conjecture: the graded distribution corresponds to a Specht module that branches successively through three rank-1 atom-carrying ancestors, while the concentrated distribution corresponds to a Specht module that has a single "cleanly inherited" atom^3 eigenvalue from one specific deep subfactor.

5.2 Surprises and structural observations

1. **Common A_9.** Both σ_{top} at $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$ have the form $q^a(1+q)^b\mathfrak{a}(q)^3A_9(q)$ with the *same* $A_9 = q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$. The palindromic core after removing atom^3 is identical across the two shapes, strongly suggesting they share a common deep cell-structure or branching ancestry. (A_9 is also the leading remainder at $V_{(3,3,1)/n=7}$, σ_3.)

2. **The leading 2 in Q_3.** At $V_{(5,2)/n=7}$, the polynomial Q_3 appearing in σ_3 has leading coefficient 2 (degree 18, palindromic, primitive over $\mathbb{Z}$). This is the only σ_k remainder across all atom^3 shapes whose primitive form has a non-unit leading coefficient. The Newton polygon argument is unaffected (the 2 is a unit in the localisation $\mathbb{Q}[q]_{(\mathfrak{a})}$). A natural guess: at degree-3 symmetric function level, two of the five eigenvalues of $\Pi_q|_{V_{(5,2)/n=7}}$ are Galois conjugates living in a quadratic extension of $\mathbb{Q}(q)$, and their joint contribution to σ_3 produces a leading 2.

3. **Splitting field.** Theorems T2 and T3 give the multiset of valuations of the roots, but do not address whether g splits over $\mathbb{Q}(q)$ or only over a finite extension. In both cases the Newton polygon is "regular" at the slope -3 (length 1): so by Newton-polygon factorisation [2, Ch. II, Cor. 6.5], g factors as $g = g_1 \cdot g_0$ in $\mathbb{Q}(q)[t]$ where g_1 is monic of degree 1 (the atom^3-carrying root, defined over $\mathbb{Q}(q)$) and g_0 has all roots of $\mathfrak{a}$-valuation 0. In particular, *the atom^3 eigenvalue is rational over* $\mathbb{Q}(q)$. The remaining roots (those of g_0) need not be: at $V_{(5,2)/n=7}$, g_0 has degree 4 and is the quartic carrying the leading-2 surprise.

4. **Sum check.** $\sum_i \nu_{\text{atom}}(e_i) = \nu_{\text{atom}}(\det(\Pi_q|_\lambda)) = \nu_{\text{atom}}(\sigma_{\text{top}})$, i.e. $3+0+0+\cdots+0 = 3$ in both cases. This is automatic from $\sigma_r = \prod e_i$ but provides a useful sanity check on the polygon argument.

5.3 Status

These are theorems in the formal sense: every step is rigorous given (i) the correctness of the symbolic Lagrange interpolation in `verify.py` (exact rational arithmetic, 89/110 evaluation points exceed the degree bounds 84/105), (ii) the standard Newton polygon theorem. No appeal to numerics or analogy is made. The conjectural content (concentrated vs. graded mechanism; A_9 as common ancestor; Galois quadratic extension at $(5,2)$) is clearly separated above.

A Coprimality verification

For each of the 9 remainder polynomials in Lemmas 4 and 5 we substitute $q = -2 + \sqrt{3}$ (a root of $\mathfrak{a} = q^2 + 4q + 1$) and verify the result is non-zero in $\mathbb{Q}(\sqrt{3})$. Equivalently, we compute the remainder of polynomial division by $\mathfrak{a}$ over $\mathbb{Z}[q]$. The script `check_coprimality.py` performs this; the table is:

Polynomial	remainder mod $\mathfrak{a}$
P_1 ($V_{(4,3)}$, σ_1)	$-28860\,q - 7733$
P_2 ($V_{(4,3)}$, σ_2)	$6{,}000{,}660\,q + 1{,}607{,}872$
P_3 ($V_{(4,3)}$, σ_3)	$4{,}378{,}860\,q + 1{,}173{,}312$
A_9 ($V_{(4,3)}$, σ_4)	$-60\,q - 16$
Q_1 ($V_{(5,2)}$, σ_1)	$-20{,}280\,q - 5{,}434$
Q_2 ($V_{(5,2)}$, σ_2)	$-1{,}135{,}260\,q - 304{,}192$
Q_3 ($V_{(5,2)}$, σ_3)	$-7{,}135{,}920\,q - 1{,}912{,}064$
Q_4 ($V_{(5,2)}$, σ_4)	$-4{,}378{,}860\,q - 1{,}173{,}312$
A_9 ($V_{(5,2)}$, σ_5)	$-60\,q - 16$

All entries are non-zero, confirming $\mathfrak{a} \nmid$ each P_k, Q_k in $\mathbb{Q}[q]$.

B Computational data summary

For reference, the explicit factorisations from the verification run:

$V_{(4,3)/n=7}$**, dim** 14**, rank** 4**:**

$$\begin{aligned}
\sigma_1 &= q^3(1+q)^3 \cdot P_1, \quad \deg P_1 = 12 \\
\sigma_2 &= q^7(1+q)^{10} \cdot P_2, \quad \deg P_2 = 18 \\
\sigma_3 &= q^{12}(1+q)^{21} \cdot P_3, \quad \deg P_3 = 18 \\
\sigma_4 &= q^{18}(1+q)^{36} \cdot \mathfrak{a}^3 \cdot A_9 \\
\sigma_5 &= 0
\end{aligned}$$

$V_{(5,2)/n=7}$**, dim** 14**, rank** 5**:**

$$\begin{aligned}
\sigma_1 &= q^2(1+q)^5 \cdot Q_1, \quad \deg Q_1 = 12 \\
\sigma_2 &= q^5(1+q)^{14} \cdot Q_2, \quad \deg Q_2 = 18 \\
\sigma_3 &= q^9(1+q)^{27} \cdot Q_3, \quad \deg Q_3 = 18, \text{ leading coef } 2 \\
\sigma_4 &= q^{13}(1+q)^{40} \cdot Q_4, \quad \deg Q_4 = 18 \\
\sigma_5 &= q^{18}(1+q)^{57} \cdot \mathfrak{a}^3 \cdot A_9 \\
\sigma_6 &= 0
\end{aligned}$$

with $A_9(q) = q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$.

References

[1] P. N. Hoefsmit, *Representations of Hecke algebras of finite groups with BN-pairs of classical types*, Ph.D. thesis, University of British Columbia, 1974.

[2] J. Neukirch, *Algebraic Number Theory*, Grundlehren der Mathematischen Wissenschaften 322, Springer-Verlag, 1999.